Interpretable Model for Predicting Seizures Using EEG Data

Al Shodiev[1], Prateek Kalakuntla[2]

[1] Depart of Biomedical Engineering; Harvard College; Cambridge, MA
[2] Department of Biological Sciences; Massachusetts Institute of Technology; Cambridge, MA

***Abstract* — In this article, we propose a novel approach in classifying raw multichannel EEG signal recordings of human patients instead of mice data from Temple University using Long-Short Term Memory (LSTM) and recurrent neural-network (RNN) classifier to detect seizures. Using exclusively the standard deviation and activity of the EEG signal, our model had the following performance measure – Area Under the Receiving Operator Characteristic Curve (AUC-ROC) of 0.91 which we found to be higher than chance and can potentially aid physicians predict whether seizures will occur with significantly higher accuracy. EEG signal recordings are complex, and detection of seizures is a challenging task, and a timely and accurate diagnosis of epilepsy is essential to identify suitable antiepileptic drug therapies. Current predicting models have AUC- ROC between 0.983 and 0.993. Although, the models have been proven to be effective, they have also been difficult to implement in a clinical setting. This is because the models are trained on mice EEG data, and they use features in the frequency domain which are difficult to interpret. Our approach is novel in that the model is trained on human data which is different from previous works.**

I. Introduction

According to National Institute of Neurological Disorders, about 2.3 million adults and more than 450,000 children and adolescents in the United States currently live with epilepsy. Each year, an estimated 150,000 people are diagnosed with epilepsy. Epilepsy affects both males and females of all races, ethnic backgrounds, and ages. In the United States alone, the annual costs associated with the epilepsies are estimated to be $15.5 billion in direct medical expenses and lost or reduced earnings and productivity [12]. Epilepsy is the condition of recurrent, unprovoked seizures. Epileptic seizure is used to distinguish a seizure caused by abnormal neuronal firing from a nonepileptic event, such as a psychogenic seizure. Epilepsy syndrome refers to a group of clinical characteristics that consistently occur together, with similar seizure type(s), age of onset, EEG findings, triggering factors, genetics, natural history, prognosis, and response to antiepileptic drug therapies [14].

Seizures can be measured using raw multichannel EEG signal recordings. EEG measures the brain signals and plays a huge role in determining whether a person has an epilepsy or has seizures caused by low blood sugar or alcohol withdrawal. These signals are complex, and detection of seizures is a challenging task. However, machine learning classifiers can classify EEG data and detect seizures. The current algorithms of classifying EEG data are time-consuming and tedious. Shoeb et al. have found Support Vector Machine (SVM), patient-specific machine learning technique, to be 96% effective in reading 173 seizures of patients from Children's Hospital Boston - MIT data using vectors as features, but the problem is that the classifier trains the dataset twice, one for SVM – Genetic Algorithm and another for SVM- Particle Swarm Optimization (PSO) which makes it time-consuming [5]. Donos et al. used time and frequency as features for random forests and reached a 93.8% on EPILAPSIAE data; the shortcoming of this technique is that the performance could decline if EEG signals are too noisy. Guo et al used ANN on BONN University data and reached a 99.6% accuracy using line length as features. Although, these models were accurate in detecting seizures, they are unable to explain their detection steps. Whereas models like random forest define each detection step which makes it human-interpretable and suitable for clinical use [6].

In another paper, Zhou et al used Convolutional Neural Networks (CNN) and found that using frequency domain signals as features, a representation of how much signal there is among each given frequency band, was more effective than using time-domain signals, a representation of signal amplitude over a time domain [3]. The CNN had only three layers, and the structure of the network was simpler than previous models. In addition, one second segments were used for detection. While this is a big step towards use in a clinical setting, extracting frequency-based features takes significantly longer than extracting time-based features thereby limiting the model's potential applications. In other works, Cho et al also used CNN and compared it to RNN and Fourier Convolutional Neural Network (FCNN) [8]. The model used raw waveform images of EEG was found to be more effective compared to the latter models. The performance results of AUC-ROC were from 0.983 to 0.984, from 0.985 to 0.989, and from 0.989 to 0.993 for FCNN, RNN, and CNN, respectively [4]. While current machine learning methods are relatively accurate, they have not been implemented in clinical settings. This is because these algorithms are not often trained on human data.

By interpreting the recorded EEG signals visually and developing an algorithm for detecting seizures, neurologists can substantially distinguish between two epileptic brain activities - during an ictal state and normal brain activities between seizures, known as interictal state. Besides that, the need for objective, rapid, and effective systems for the processing of vast amounts of EEG recordings has become unavoidable to be able to diminish the possibility of misinterpretations. The availability of such systems would greatly enhance the quality of life of epileptic patients.

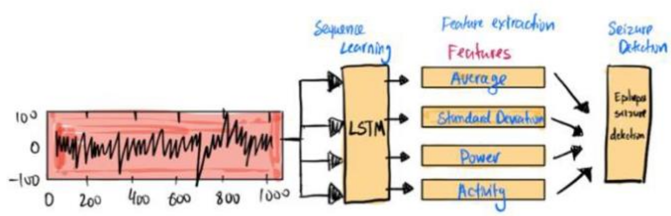


Figure 1: An illustration of EEG-based seizure detection system

II. **Methods**

For this paper, all the data was obtained from the Temple University EEG Seizure Data Corpus [9]. The subset of the data that was used contains data from 264 patients over 579 sessions. The patients experienced 1327 seizures lasting a total of 25 hours. The seizures were of four main types, but we chose to group them together for our analysis. Given this fact, we chose to up-sample the sessions containing seizures such that seizure events made up, on average, 33% of the input dataset. The seizure data was taken using a 10-20 EEG configuration and contains 21 channels of EEG signal recordings. The start and stop time, as designated by an attending physician, of each seizure is also labeled. We randomized the training and test sets. In both the classification and prediction tasks, we used 80% of the data for training and 20% for validation.

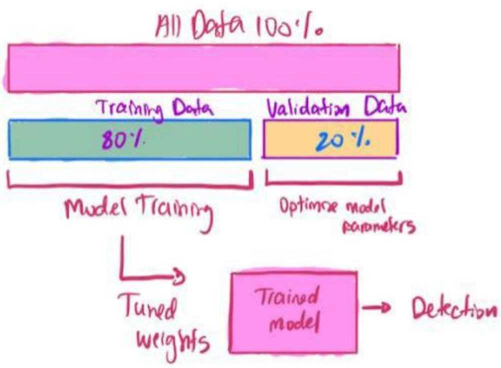


Figure 2: Sketch of Data Partitioning used to create the model

Most current models use raw EEG data for seizure prediction, but we extracted specific features from the EEG data which we will mention in a bit. First, we processed the EEG signal recordings by separating each session into bins ranging from half a second to ten seconds. We then labeled these bins based on whether the patient was undergoing a seizure at any point during the bin and extracted a series of time-based features from each channel. We chose to extract 4 features - the average, the standard deviation, the power, and the activity. The power of an EEG signal is a measure of the absolute distance of the signal from 0, while the activity represents the speed at which the system oscillates as a function of power. Next, we calculated each of these features for each of the 21 EEG channels, resulting in 84 features that were passed into the model at each time point [7]. We chose the aforementioned features because we wanted our model to be constrained to features that are human-interpretable. This method allows us to develop an easily interpretable model that can be used by neurologists or physicians in a clinical setting.

III. **Results and Discussion**

Classification Model

Since EEG Data is sequential and there is no inherent spatial organization to the features, LSTM is the best algorithm to use in our case. We found the following parameters in our algorithm to most accurate- a 2-layer LSTM each with 100 outputs, a learning rate of .001, and a lambda of 1e-07.
To find the optimal model parameters, we adopted a simple grid search strategy. We limited the search to a few variables per parameter because there were too many to perform an exhaustive search. We searched the number of outputs of the LSTM layer (Supplemental Figure 1), the depth of the network (Supplemental Figure 2), the network learning rate (Supplemental Figure 3), and the L2 lambda (Supplemental Figure 4). None of these metrics affected the model's accuracy so we selected these parameters because they were less time - consuming per epoch (Supplemental Figure 4).

*Note: Still need to upload SF 3 and 4

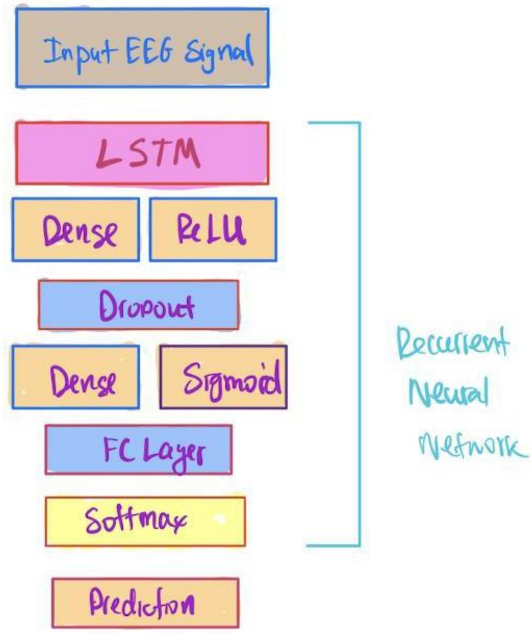


Figure 3: An Overview Sketch of the Proposed Architecture

The goal of our model was to classify whether a seizure occurred during the bin. Based on Figure 4, the model's accuracy increases as we decrease the bin size.

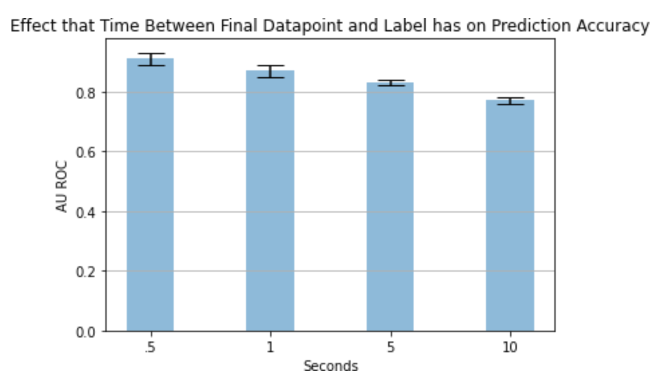


Figure 4: Prediction results of the model with different bin sizes. Each experiment was completed 20 times to get error bars, as were all future experiments. AUC for half second bins was .91 ± .02, second bins was .87 ± .02, five second bins was .83 ± .01, and ten second bins was .77 ± .01

There are a few explanations for why model's accuracy increases as time decreases. First, we determined that the threshold for accurate results is 0.8 or more AUC – ROC. As the amount of time between the final datapoint and the prediction was smaller, the model was given more relevant information before it needed to identify whether the next bin contained a seizure or not. This explanation must be investigated further for confirmation. We could determine where the most valuable seizure markers are by running our model with increasingly smaller bin sizes and identifying when the accuracy stops increasing. This demonstrates one of the limitations of the model. Specifically, the further away one is from the seizure itself, the harder it is to predict whether it will occur. The drastic drop-off in accuracy between 1 second bins and 10 second bins implies that accurate predictions of seizure state, with the features that we are using, cannot be made more than 10 seconds before the seizure occurs.

Our current model has an AU ROC of .91 as depicted in figure 5. While this AU ROC is not as high as LSTM standard, .985 in *Cho et al*, it is significantly higher than chance. Given the limited input space, this model could help doctors predict a seizure with significantly high accuracy.

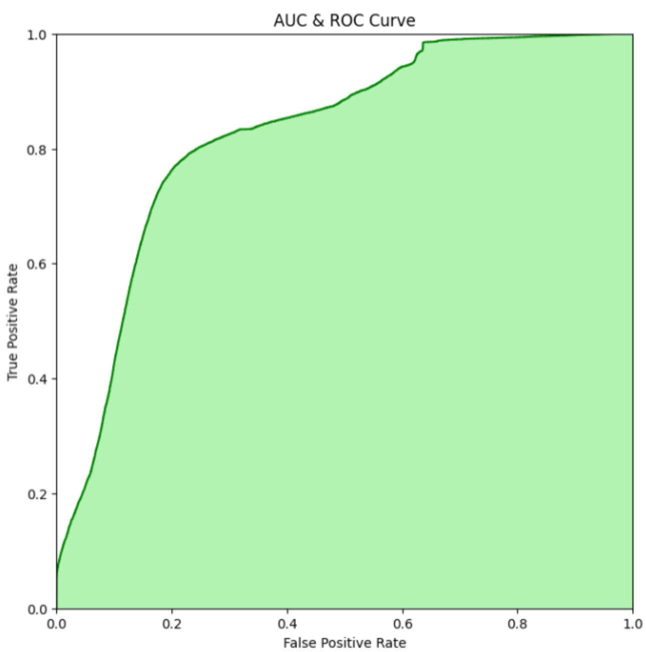


Figure 5: AU ROC graph for model with half second bins, 2 100 output LSTM layers, a learning rate of .001, and a lambda of 1e-07. The area under the curve is .91.

To analyze the information provided by each of the features - average, standard deviation, activity, and power - we decided to test the model's accuracy by allowing it access to only one feature from each of the 21 channels. The results of this experiment are depicted in figure 3. The motivation behind learning what each feature does is to further simplify the model and understand which feature plays a more significant role in detecting seizures [15][16]. The objective of a seizure detection system is to distinguish the seizure state from the seizure-free state. The input of such systems can be two types - a feature, or a group of features extracted from EEG signals. Feature extraction step consists of three steps - preprocessing, feature computation and feature reduction which involve a considerable amount of computation [16]. We need to optimize this step for activity and power to be applicable in a real-time seizure detection system. In order to have such efficient detection system in terms of activity and power while not losing reliability, the most informative features should be determined out of a pool of features which is why we analyze the performance of each individual feature from the EEG Dataset.

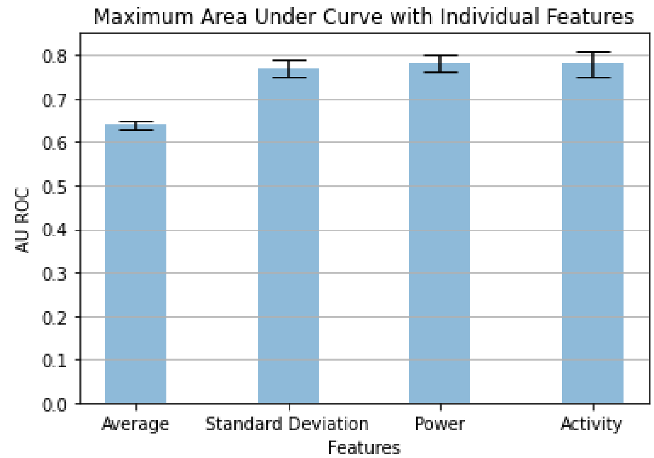


Figure 6: ROC AUC when the model was trained and validated with a single feature. Model parameters: half second bins, 2 100 output LSTM layers, a learning rate of .001, and a lambda of 1e-07. The AUC of the "average" model was .64 ± .01, the AUC of the "standard deviation" model was .77 ± .02, the AUC of the "power" model was .78 ± .02, and the AUC of the "activity" model was .79 ± .03

The figure clearly shows two facts. First, that the model's accuracy increases dramatically when it can analyze multiple features. Second, the average value of the EEG signal gave the least information, while the standard deviation, power, and activity all gave a similar amount of information. We have a few hypotheses as to why the average value of the EEG signal gave so little information to our model. First, as has been shown throughout the literature but specifically by *Subha et al*, EEG baselines vary greatly between people. As such, an average that represents one person undergoing a seizure may represent baseline for another, very similar, person. However, this alone does not explain why the power of the signal, which can also be thought of as a time averaged measurement, has so much more information than the average. The most likely reason for this is the large amount of variability in the EEG data [11]. Within a single channel over the course of microseconds, the signal can jump from a large positive value to a large negative value, or from a small positive to small negative value. Taking the average of the signal washes out all this information because a bin with large positive and negative signals and a bin with small positive and negative signals will look the same but taking the power of the signal retains this information. Others have tried similar approach – extracting features from EEG signal recordings to examine their significance [15]. *Niknazar et al*. analyzed 17 different features on 88 seizures which were Bhattacharyya-based dissimilarity index (BBDI), Approximate Entropy, Higuchi fractal dimension, Lyapunov exponent and more. and found that BBDI had the highest statistical significance both within seizure-onset zone and outside seizure-onset zone [13].

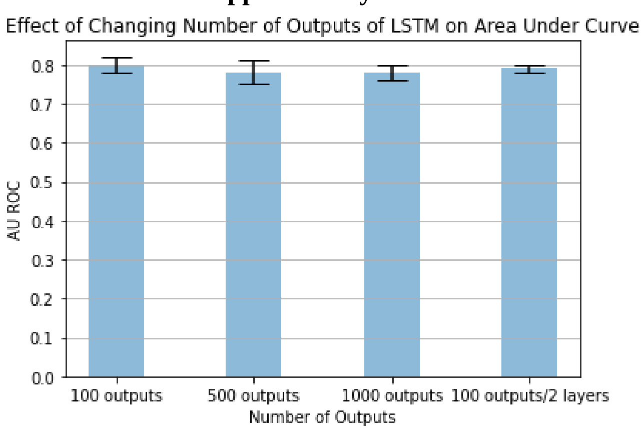
Our data shows that this information about the intensity of the signal, regardless of whether the brain is producing a positive or negative voltage, is valuable for modeling whether a subject is about to undergo a seizure. Evaluating the performance of each individual feature is important because we can show that a certain feature, in this case power, with a simple thresholding detection method can achieve comparable results compared to studies which use complex detection methods like *Shoeb et al* [5].

IV. **Conclusion**

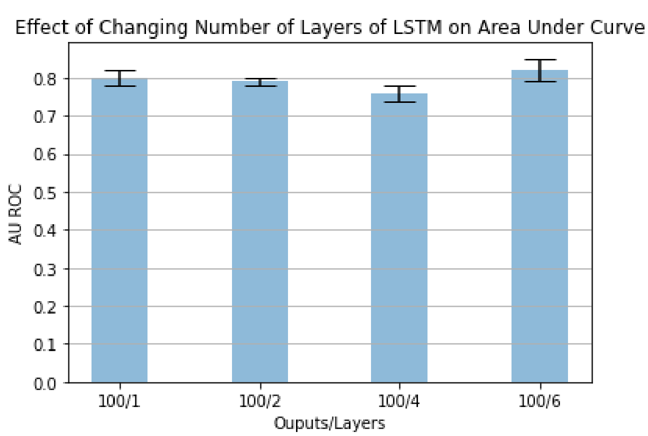
In this paper, we were able to use EEG data to develop a model that can be used to predict whether a patient will undergo a seizure. This model uses an artificially limited feature space which only includes easily interpretable features. This will make it easier to extract relevant heuristics to provide to physicians so that they can predict seizures in real time, which will be investigated in the future. To do this, we plan to use backpropagation and weight visualization to interpret the features the model is looking for to make its predictions, and what features maximally activate the model. We will then use these interpretations to develop heuristics for predicting seizures that we can give to physicians.

Moreover, in future studies, we can shift our focus and predict the seizure duration. First, we would try to thoroughly understand whether the amount of information needed to predict seizure duration is present in the EEG data alone, and whether other non-invasive methods of neural analysis including fMRI and functional interferometric diffusing wave spectroscopy, could help us get enough information to develop a clinically viable model.

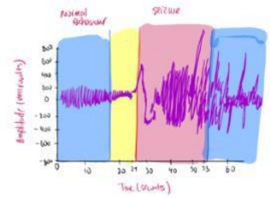
V. **Supplementary Information**



Supplemental Figure 1



Supplemental Figure 2



Supplemental Figure 3
Normal State vs Seizure State

VI. **References**

1. Sperling MR. The consequences of uncontrolled epilepsy. CNS Spectr. 2004 Feb;9(2):98-101, 106-9. doi: 10.1017/s1092852900008464. PMID: 14999166.
2. Kumar, Yatindra, et al. “Relative Wavelet Energy and Wavelet Entropy Based Epileptic Brain Signals Classification.” *Biomedical Engineering Letters*, vol. 2, no. 3, Sept. 2012, pp. 147–157, link.springer.com/article/10.1007/s13534-012-0066-7, 10.1007/s13534-012-0066-7. Accessed March 16. 2022.
3.Zhou M, Tian C, Cao R, Wang B, Niu Y, Hu T, Guo H, Xiang J. Epileptic Seizure Detection Based on EEG Signals and CNN. Front Neuroinform. 2018 Dec 10;12:95. doi: 10.3389/fninf.2018.00095. PMID: 30618700; PMCID: PMC6295451.
4.Siddiqui, Mohammad Khubeb et al. “A review of epileptic seizure detection using machine learning classifiers.” Brain informatics vol. 7,1 5. 25 May. 2020, doi:10.1186/s40708-020-00105-1
5.Shoeb A, Guttag J (2010) Application of machine learning to epileptic seizure detection. In: 2010 the 27th International Conference on Machinelearning, Haifa, Israel
6.Donos C, Dümpelmann M, Schulze-Bonhage A. Early seizure detection algorithm based on intracranial EEG and random forest classification. Int J Neur Syst. 2015;25(05):1550023.
7.Guo L, Rivero D, Dorado J, Rabunal JR, Pazos A. Automatic epileptic seizure detection in eegs based on line length feature and artificial neural networks. J Neurosci Methods. 2010;191(1):101–109.
8.Cho, KO., Jang, HJ. Comparison of different input modalities and network structures for deep learning-based seizure detection. Sci Rep 10, 122 (2020). https://doi.org/10.1038/s41598-019-56958-y
9.Temple University Data
10.Liang, Z., Oba, S., &; Ishii, S. (2019). An unsupervised EEG decoding system for human emotion recognition. Neural Networks, 116, 257-268. doi:10.1016/j.neunet.2019.04.003
11.Puttanakil, S., Joesph, P., & Acharya, R. (2010). EEG Signal Analysis: A Survey. Journal of Medical Systems, 34, 195-212.
12.“The Epilepsies and Seizures: Hope through Research | National Institute of Neurological Disorders and Stroke.” Nih.gov, 2018, www.ninds.nih.gov/Disorders/Patient-Caregiver-Education/Hope-Through-Research/Epilepsies-and-Seizures-Hope-Through#3109_2. Accessed March 28. 2022.
13.Niknazar, Hamid, et al. “Performance Analysis of EEG Seizure Detection Features.” Epilepsy Research, vol. 167, Nov. 2020, p. 106483, www.sciencedirect.com/science/article/pii/S0920121120305349, 10.1016/j.eplepsyres.2020.106483. Accessed March 28. 2022.
14.Stafstrom, C. E., and L. Carmant. “Seizures and Epilepsy: An Overview for Neuroscientists.” Cold Spring Harbor Perspectives in Medicine, vol. 5, no. 6,1June2015,pp.a022426–a022426, www.ncbi.nlm.nih.gov/pmc/articles/PMC4448698/#A022426C23, 10.1101/cshperspect.a022426. Accessed March 28. 2022.
15.Ma, Qingguo, et al. “A Novel Recurrent Neural Network to Classify EEG Signals for Customers’ Decision-Making Behavior Prediction in Brand Extension Scenario.” Frontiers in Human Neuroscience, vol. 15, 8 Mar. 2021, www.frontiersin.org/articles/10.3389/fnhum.2021.610890/full#:~:text=By%20combining%20the%20t%2DSNE,and%20test%20accuracy%20was%2088.32%25., 10.3389/fnhum.2021.610890. Accessed March 28. 2022.
16.Abdelhameed, Ahmed, and Magdy Bayoumi. “A Deep Learning Approach for Automatic Seizure Detection in Children with Epilepsy.” Frontiers in Computational Neuroscience, vol. 15, 8 Apr. 2021, www.frontiersin.org/articles/10.3389/fncom.2021.650050/full, 10.3389/fncom.2021.650050. Accessed March 28. 2022.

VII. **Contributions**

PK and AS built majority of the dataset. AS organized the data and did some data cleaning. PK also cleaned the data. PK built the model; AS and PK planned which model to use for this case. AS wrote the methods, analyzed results and discussed them. AS wrote the conclusion, introduction, and abstract, generated figures.